

COMM144 Generative AI

1st Workshop

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Exercise 1

Consider a simple poker game. Suppose a deck of poker cards contains four suits: Clubs, Spades, Hearts, and Diamonds, with each suit having 10 cards numbered from 1 to 10. So the total number of cards is 40. The card with the number 10 is the largest, while the card with the number 1 is the smallest. Cards of different suits with the same number are considered equal.

- Suppose person A fetches a card with the number 3 of Clubs from the deck. Person B then fetches another card from the deck. What is the probability that person B's card is less than person A's card?
- If you have drawn four cards sequentially from the deck of 40 cards. What is the total probability that the four cards are of Clubs?
- If you have drawn four cards sequentially from the deck of 40 cards. What is the total probability that the four cards have the same number 5?

Exercise 2

Use the inequality $\log(x) \leq x - 1$ to show that the KL divergence is nonnegative, where it is defined as $D_{KL}(P||Q) = \int p(\mathbf{x}) \log \left(\frac{p(\mathbf{x})}{q(\mathbf{x})} \right) d\mathbf{x}$.

Under what condition does the KL divergence equal to 0?

Exercise 3

Suppose X is a binary discrete random variable with $P_X(x = 1) = 0.8$ and $P_X(x = 0) = 0.2$.

Suppose Y is another binary discrete random variable with $P_Y(y = 1) = 0.4$ and $P_Y(y = 0) = 0.6$.

- What is the entropy $H(X)$?
- What is the cross-entropy $H(X, Y)$?
- What is the KL divergence $D_{KL}(X||Y)$?

Exercise 4

See the questions within the python notebook "COMM114_w1.ipynb".

1. Open the notebook by using Google Colab.
2. Run the sample python code to compute Entropy and KL-divergence.

Exercise 5

Read the article titled "Entropy, Cross-Entropy, and KL Divergence: Mathematical Foundations and Applications" via [the link](#)